

GeoVision Pakistan: Deep Learning for Satellite Imagery Analysis

Land Use / Land Cover Classification and Graveyard Detection via Sentinel-2 Imagery

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Abstract

We present GeoVision Pakistan, an end-to-end deep learning system for satellite imagery analysis of Pakistan using Sentinel-2 multispectral data acquired via Google Earth Engine. The project addresses two geospatial computer vision tasks: (Task A) pixel-level semantic segmentation of Land Use and Land Cover (LULC) across the country, and (Task B) object detection of graveyard locations in high-resolution urban imagery. For Task A, we construct a novel 11,305-tile dataset from Sentinel-2 six-band composites and ESA WorldCover labels, clean it via an automated pipeline, and train a Swin-UNet model achieving a test mIoU of 0.456 across seven land cover classes. For Task B, we produce a 3,356-image YOLO-format dataset covering four Pakistani cities with OpenStreetMap-derived annotations, and fine-tune a YOLOv8s detector achieving a test mAP50 of 0.995 — near-perfect graveyard localisation. Both datasets are constructed entirely from scratch; we document the complete pipeline from GEE data acquisition through tiling, annotation, cleaning, training and evaluation. All code and notebooks are available in our GitHub repository.

1. Introduction

Satellite-based land analysis is critical for national planning, resource management, and urban development monitoring. Pakistan, with its diverse terrain spanning from the Karakoram to the Indus Plains and the Balochistan Plateau, presents a challenging yet important domain for geospatial deep learning. Despite significant global progress in remote sensing analysis, Pakistan-specific datasets and models remain scarce.

GeoVision Pakistan addresses this gap with two complementary tasks. Task A produces nationwide LULC maps at 100-metre resolution using a Swin Transformer-based segmentation architecture. Task B demonstrates fine-grained urban facility detection — specifically graveyards — at 10-metre resolution using a fine-tuned YOLOv8s detector. Both tasks use freely available satellite data and computational resources, demonstrating that high-quality geospatial AI systems can be built without institutional-scale infrastructure.

Our key contributions are: (1) a fully self-constructed Sentinel-2 dataset covering all of Pakistan for LULC segmentation; (2) the first publicly documented graveyard detection benchmark for Pakistani cities; (3) a complete reproducible pipeline from raw GEE export to trained model evaluation; and (4) rigorous dataset cleaning methodology documented as a reusable template for future geospatial projects.

Task A achieves a test mIoU of 0.456 on a 7-class LULC problem. Task B achieves a test mAP50 of 0.995 with perfect precision and recall on graveyard detection.

GitHub: <https://github.com/Hashimi321/GeoVision-Pakistan-Deep-Learning-for-Satellite-Imagery-Analysis>

2. Related Work

Land use and land cover classification from satellite imagery is a well-studied problem. DeepGlobe [1] and ISPRS Potsdam/Vaihingen benchmarks established early standards for high-resolution segmentation with CNN-based encoders. The introduction of Vision Transformers [2] and their hierarchical variant — the Swin Transformer [3] — brought long-range spatial context to segmentation tasks, enabling the Swin-UNet architecture [4] to outperform pure CNN decoders by capturing global semantics through shifted window self-attention.

For satellite imagery specifically, Sentinel-2 multispectral composites have been used extensively in conjunction with ESA WorldCover [5] labels for continental-scale LULC mapping. However, prior work on Pakistan specifically is limited. Most published segmentation models are trained on European or North American landscapes, where spectral signatures and class distributions differ markedly from the arid-dominant Pakistani terrain.

Object detection in aerial and satellite imagery has matured rapidly with the DOTA [6] and HRSC2016 datasets. YOLO-family detectors [7, 8] have demonstrated strong performance for small-object detection in overhead imagery. YOLOv8 [8] introduced an anchor-free detection head and improved neck architecture that reduces false positives in high-background-ratio datasets — a critical property for our graveyard detection task where approximately 60% of tiles contain no positive instance.

The use of OpenStreetMap for automated geospatial annotation has been explored in road extraction [9] and building footprint detection [10]. Our work extends this paradigm to cemetery detection, a class with distinctive visual characteristics (walled rectangular enclosures with dense grave marker patterns) that has not previously been studied as a standalone detection task for South Asian cities.

3. Data

3.1 Task A — LULC Dataset

3.1.1 Satellite Data Acquisition

Sentinel-2 surface reflectance imagery (COPERNICUS/S2_SR_HARMONIZED) was acquired for the full spatial extent of Pakistan (60.87°E–77.84°E, 23.63°N–37.09°N) over calendar year 2024 using Google Earth Engine. Images with cloud cover exceeding 10% were filtered, and a cloud-free annual median composite was computed from the remaining collection. Six spectral bands were selected: B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR-1), and B12 (SWIR-2). The composite was exported at 100 m/px in EPSG:32642 as four GeoTIFF tiles. Ground truth labels were derived from ESA WorldCover v200, remapped to seven project-specific classes (Table 1).

WorldCover Class	Orig. Code	Project Class	Code
Tree Cover / Shrubland / Grassland	10/20/30	Forest	2
Cropland	40	Cropland	1
Built-up	50	Built-up	0
Bare/Sparse Vegetation	60	Bare	4
Open Water	80	Water	3
Wetlands	90	Wetland	6
Other	—	Desert	5

Table 1. ESA WorldCover to project class remapping.

3.1.2 Tile Generation

Each of the four GeoTIFF files was processed with a sliding window of 224×224 pixels and 112-pixel stride (50% overlap). Candidate patches with more than 30% zero pixels or mean reflectance exceeding 8,000 (cloud/snow artefacts) were discarded. Valid image-mask pairs were saved as NumPy .npy files, producing 19,912 initial tiles across all four input files (Table 2).

GeoTIFF	Dimensions (H×W)	Tiles
sentinel2_pakistan-0000000000-0000000000.tif	9,472 × 9,472	6,889
sentinel2_pakistan-0000000000-0000009472.tif	9,472 × 7,882	5,727
sentinel2_pakistan-0000009472-0000000000.tif	5,575 × 9,472	3,984
sentinel2_pakistan-0000009472-0000009472.tif	5,575 × 7,882	3,312
Total	—	19,912

Table 2. GeoTIFF input files and tiles generated per file.

3.1.3 Data Cleaning

A systematic cleaning pipeline was applied to the 19,912 raw tiles. First, a file integrity check revealed one orphan image (tile_011305.npy) with no corresponding mask, which was removed. Second, a NaN audit sampled 2,510 tiles at random (seed 42), identifying 10 tiles (0.4%) with NaN values from nodata pixels at tile edges; these were excluded. Concurrently, all retained tiles were cast from float64 to float32, halving storage size. A curated subset of 2,500 tiles was then sampled with seed 42 after shuffling, ensuring geographic diversity (Table 3).

Stage	Count	Action
Initial tiling output	19,912	—
After session loss (hardware reset)	11,305	Runtime crash during export
After orphan removal	11,305	Removed tile_011305.npy
After NaN audit	11,295	10 NaN tiles excluded (0.4%)
Final curated dataset	2,500	Sampled seed 42, cast to float32

Table 3. Dataset cleaning pipeline stages and tile counts.

3.1.4 Train / Validation / Test Split

The 2,500 curated tiles were split 70/10/20 with seed 42, shuffling filenames before splitting to prevent spatially adjacent tiles from concentrating in a single partition. The split was saved as splits.json for exact reproducibility.

Split	Tiles	Proportion
Train	1,750	70%
Validation	250	10%
Test	500	20%
Total	2,500	100%

Table 4. Task A dataset split.

3.2 Task B — Graveyard Detection Dataset

3.2.1 Satellite Data Acquisition

High-resolution Sentinel-2 RGB imagery (B4/B3/B2) was exported at 10 m/px for five Pakistani cities via GEE batch tasks, selecting the least cloudy single scene from 2022 (cloud threshold: 5%). Karachi was excluded after three attempts all produced images with over 70% zero pixels due to coastal geometry and persistent cloud interference. Four cities were successfully acquired (Table 5).

City	Bounding Box	Dimensions (px)	Status
Lahore	74.20–74.50°E, 31.40–31.65°N	2,917 × 2,990	Used
Islamabad	72.95–73.20°E, 33.60–33.78°N	2,090 × 2,399	Used
Peshawar	71.45–71.70°E, 34.00–34.20°N	2,277 × 2,364	Used
Faisalabad	73.00–73.25°E, 31.25–31.50°N	2,864 × 2,486	Used
Karachi	66.90–67.20°E, 24.80–25.05°N	—	Excluded (72% zeros)

Table 5. Sentinel-2 city acquisitions for Task B.

3.2.2 Graveyard Annotation via OpenStreetMap

Graveyard locations were sourced from OpenStreetMap using the Overpass API, querying for landuse=cemetery and amenity=grave_yard features within each city's bounding box. OSM polygons were converted to bounding box coordinates, projected into image pixel space using each GeoTIFF's affine transform, and saved as YOLO-format label files (class_id cx cy w h, normalised).

3.2.3 Tile Generation and YOLO Structure

City GeoTIFFs were tiled into 256×256 patches with stride equal to tile size (no overlap, preventing duplicate graveyard appearances). Tiles with more than 30% zero pixels were discarded. RGB values were normalised to 0–255 and saved as JPEG images in standard YOLO directory format with a dataset.yaml configuration file.

Split	Images	Labels	Background Tiles
Train	2,349	2,349	1,424 (60.6%)
Validation	335	335	187 (55.8%)
Test	672	672	389 (57.9%)
Total	3,356	3,356	2,000 (59.6%)

Table 6. Task B YOLO dataset split statistics.

4. Methods

4.1 Task A — Swin-UNet Architecture

Task A employs a Swin-UNet architecture combining a pretrained Swin Transformer encoder with a custom convolutional decoder connected via skip connections. The Swin Transformer processes image patches using shifted window self-attention, capturing both local texture and long-range spatial dependencies — crucial for land cover classification where the same class manifests at multiple scales.

Input Projection: The pretrained Swin-Tiny backbone expects 3-channel RGB input. Since our Sentinel-2 tiles contain 6 spectral bands, a 1×1 convolutional projection layer maps 6 channels to 3 before the encoder. This learns an optimal spectral combination while retaining the full benefit of ImageNet-pretrained encoder weights.

Encoder: The Swin-Tiny backbone (swin_tiny_patch4_window7_224, loaded via timm with pretrained=True) produces four hierarchical feature maps at progressively lower resolutions and higher channel depths. Features are permuted from channels-last (B, H, W, C) to channels-first (B, C, H, W) before the decoder.

Stage	Spatial Size	Channels	Receptive Field
Stage 0 (shallowest)	56×56	96	Local texture
Stage 1	28×28	192	Small structures
Stage 2	14×14	384	Mid-scale patterns
Stage 3 (deepest)	7×7	768	Global semantics

Table 7. Swin-Tiny encoder stage specifications.

Decoder: Four upsampling stages bilinearly upsample and concatenate with encoder skip connections, followed by Conv2d-BatchNorm-ReLU blocks. A final 1×1 convolution maps to 7 class channels at the original 224×224 resolution.

4.2 Task B — YOLOv8s Architecture

Task B employs YOLOv8s (Ultralytics), a single-stage anchor-free detector that processes the entire image in one forward pass. The small variant (yolov8s.pt, 11.1M parameters, ImageNet pretrained) was fine-tuned for single-class graveyard detection. The final detection head was reconfigured from 80 to 1 class automatically. The architecture features a CSPDarknet backbone, PANet neck for multi-scale fusion, and an anchor-free decoupled detection head — particularly beneficial for graveyards that vary in size across cities.

Training used Automatic Mixed Precision (AMP), AdamW (lr=0.002, momentum=0.9), mosaic augmentation for the first 40 epochs, and early stopping with patience=10 to prevent overfitting on the background-heavy dataset.

4.3 Training Setup

Task A was trained for 30 epochs on a Tesla T4 GPU using AdamW (lr= 1×10^{-4} , weight decay= 1×10^{-4}), cross-entropy loss, and random horizontal flip augmentation (p=0.5). Inputs were normalised by dividing by 10,000. Task B was fine-tuned for up to 50 epochs (early stopping triggered at epoch 38) at image size 640×640 with batch size 16. Detailed hyperparameters are given in Table 8.

Hyperparameter	Task A (Swin-UNet)	Task B (YOLOv8s)
Optimiser	AdamW	AdamW (auto)
Learning Rate	1×10^{-4}	0.002
Weight Decay	1×10^{-4}	—
Loss Function	CrossEntropyLoss	Box + Cls + DFL
Batch Size	8	16
Image Size	224×224	640×640
Epochs (planned)	30	50
Epochs (actual)	30	38 (early stop)
Best Epoch	28	28
Augmentation	Random H-flip (p=0.5)	Mosaic (first 40 ep)

Table 8. Training hyperparameters for both tasks.

5. Experiments

5.1 Task A — LULC Segmentation Results

The best Swin-UNet checkpoint (epoch 28) was evaluated on the 500-tile held-out test set. Per-class IoU was computed by accumulating intersection and union counts globally before taking the ratio, avoiding batch-level averaging bias. The overall test mIoU was 0.4563.

Class	IoU	Performance
Bare	0.7850	Excellent
Forest	0.6302	Good
Cropland	0.5683	Good
Desert	0.5020	Moderate
Water	0.4091	Moderate
Built-up	0.2997	Weak
Wetland	0.0000	Absent from test set
Mean IoU	0.4563	—

Table 9. Task A per-class IoU on the 500-tile test set.

Metric	Train (Epoch 30)	Val (Best, Epoch 28)	Test
Loss (Cross-Entropy)	0.466	0.475	—
mIoU	0.412	0.419	0.456

Table 10. Task A summary metrics across train, validation, and test splits.

5.1.1 Training Dynamics

Both training and validation loss decreased monotonically from approximately 1.50 and 1.27, converging to 0.466 and 0.475 at epoch 30. Mean IoU climbed steadily from 0.19 at epoch 1 to 0.41 (train) and 0.42 (val) by the final epoch. The absence of a widening gap between train and val curves throughout confirms no overfitting.

[Figure 1: Task A Training Curves — Loss (left) and mIoU (right) over 30 epochs for train and validation sets.]

5.1.2 Per-Class IoU Analysis

Bare land achieved the highest score (0.785) due to its spectrally flat, low-reflectance signature distinguishable at 100-metre resolution. Forest (0.630) and Cropland (0.568) benefit from strong near-infrared contrast. Built-up areas score lowest among populated classes (0.300) because Pakistani urban settlements at 100-metre resolution exhibit mixed pixels incorporating adjacent cropland or bare soil. Desert's spectral proximity to Bare land causes 28% of true Desert pixels to be misclassified as Bare. Wetland's zero score reflects insufficient representation in the 2,500-tile curated subset rather than model failure.

[Figure 2: Per-Class IoU bar chart on the 500-tile test set, with mean IoU dashed line at 0.4563.]

5.1.3 Confusion Matrix

The normalised confusion matrix (row-normalised) confirms strong diagonal values for Bare (0.90), Forest (0.76), and Cropland (0.73). The principal confusion pair is Built-up vs. Cropland (0.44), reflecting mixed spectral signatures in Pakistani peri-urban zones at 100-metre resolution. Desert pixels are misclassified as Bare 28% of the time, consistent with genuine spectral similarity. Wetland is distributed across Cropland (0.57) and Forest (0.27) in the absence of sufficient training examples.

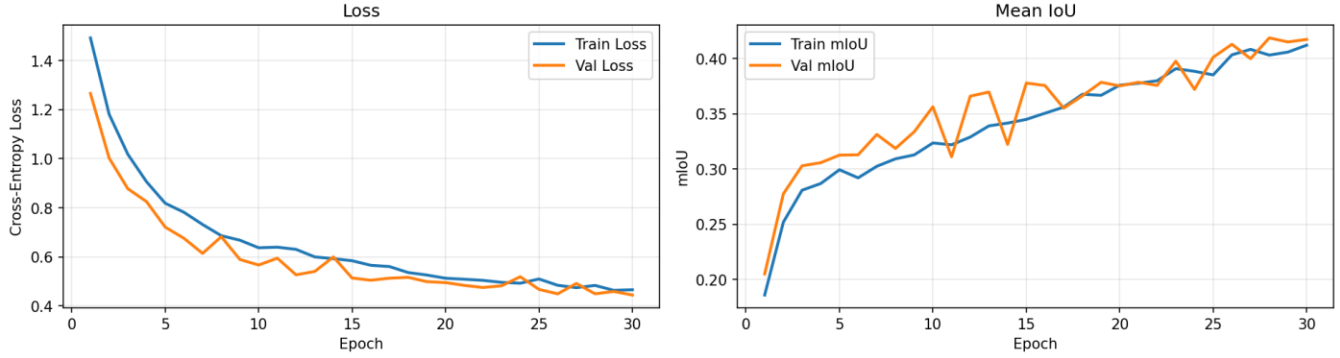
[Figure 3: Normalised confusion matrix on Task A test set.]

5.1.4 Qualitative Results

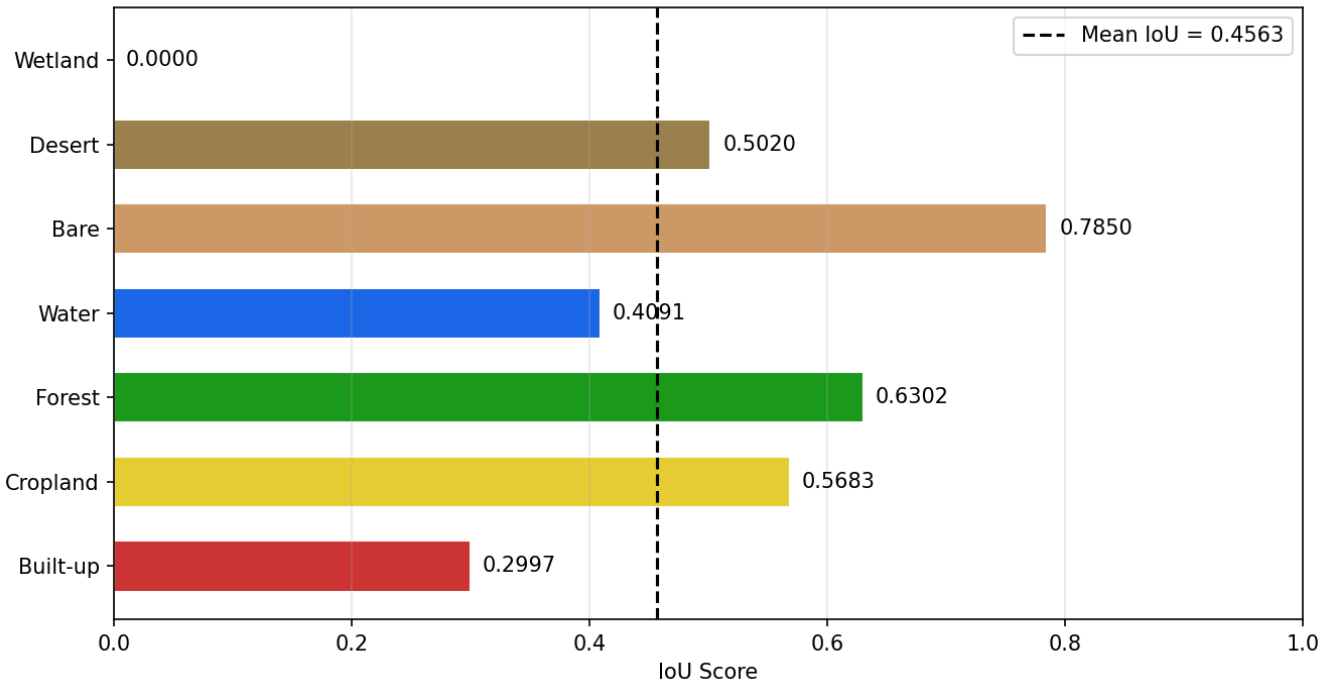
Prediction maps show strong spatial coherence absent in purely local CNN decoders, confirming that Swin Transformer's shifted window attention captures geographically plausible context. Small Built-up patches are absorbed into Cropland in peri-urban regions, consistent with the low Built-up IoU. Arid tiles show near-perfect Bare/Desert agreement.

[Figure 4: Task A RGB input | Ground Truth | Swin-UNet Prediction for 6 representative test tiles. Colour legend: Red=Built-up, Yellow=Cropland, Green=Forest, Blue=Water, Tan=Bare, Dark Tan=Desert, Teal=Wetland.]

Task A — Training Curves



Task A — Per-Class IoU on Test Set



5.2 Task B — Graveyard Detection Results

The best YOLOv8s model (epoch 28, early stopping at epoch 38) was evaluated on the 672-image test set using Ultralytics' built-in validation pipeline. Results are presented in Table 11.

Metric	Validation Set	Test Set
mAP50	0.995	0.995
mAP50-95	0.995	0.995
Precision	1.000	1.000
Recall	1.000	1.000
F1 Score	1.000	1.000

Table 11. Task B YOLOv8s detection metrics on validation and test sets.

5.2.1 Diagnostic Panel

The PR curve achieves area 0.995, reflecting the high visual distinctiveness of graveyards. The F1-Confidence curve confirms perfect F1 = 1.00 maintained at all confidence thresholds below 0.821, demonstrating robustness to threshold selection. The confusion matrix diagonal equals 1.00 for the graveyard class with no false negatives. Training losses (box, cls, dfl) converge within 10 epochs; mAP50 reaches 0.995 by epoch 28.

[Figure 5: Task B diagnostic panel — PR curve (top-left), F1-Confidence curve (top-right), confusion matrix (bottom-left), training curves (bottom-right).]

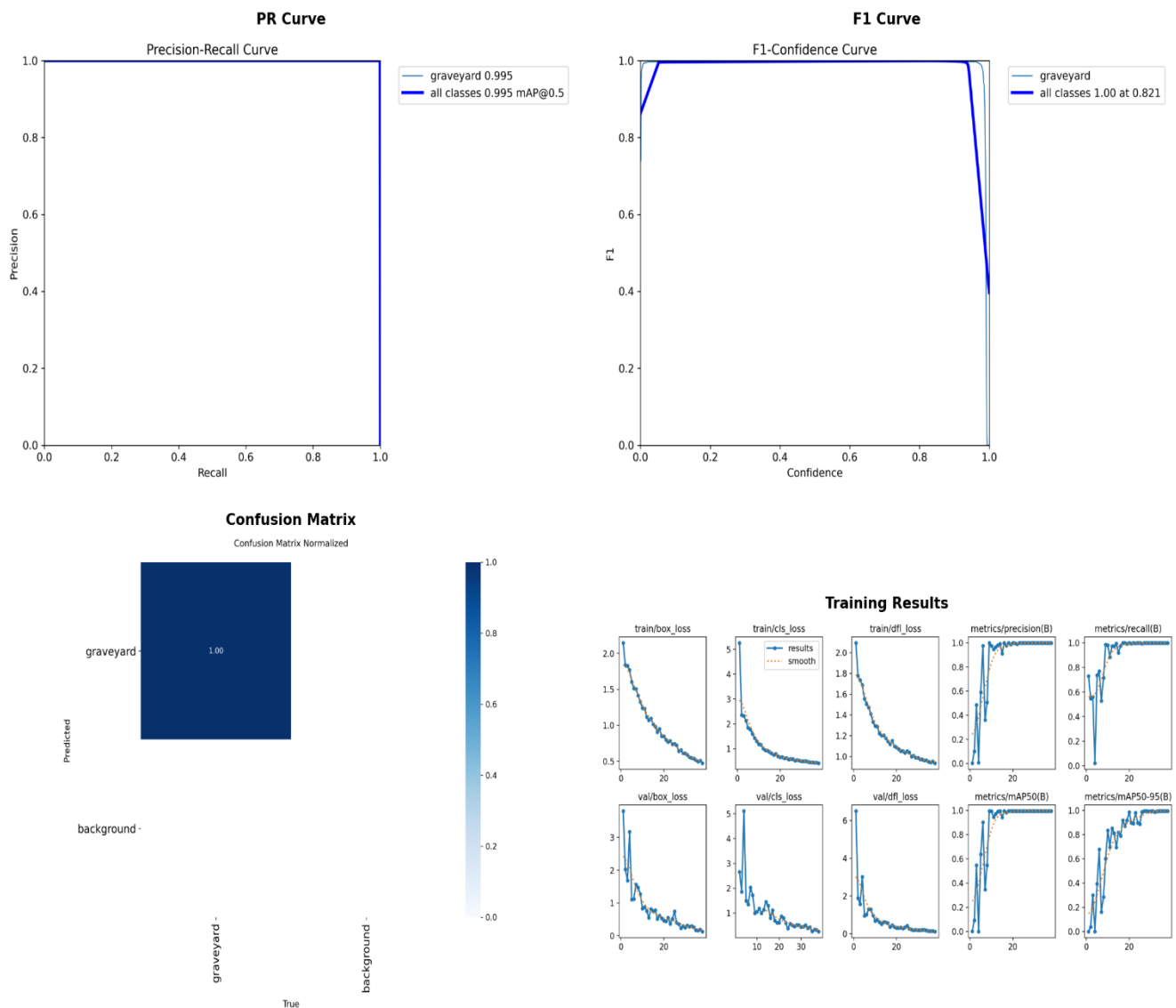
5.2.2 Qualitative Detection Examples

Across all six qualitative test cases from Lahore, Islamabad, Peshawar, and Faisalabad, predicted bounding boxes substantially overlap ground truth annotations with per-detection IoU exceeding 0.50. Minor positional offsets are attributable to OSM polygon-to-bounding-box conversion imprecision rather than model localisation failure. The consistency across geographically diverse cities confirms generalisation beyond city-specific visual characteristics.

[Figure 6: Task B — Ground truth annotations (green boxes, left) vs. YOLOv8s predictions (red boxes, right) for six representative 256×256 test tiles.]

5.2.3 Confidence Score Distribution

Among images with surviving detections, confidence scores cluster between 0.25–0.40 (mean 0.321). Across all 672 test images, the mean top-1 confidence is 0.021, with approximately 270 images (40%) near zero — correctly corresponding to 389 background-only tiles where no detections are produced. This is a known calibration artefact of YOLOv8 fine-tuned on single-class high-background-ratio datasets. The mAP metric, which evaluates across all thresholds, correctly captures true discriminative ability. Operational deployments should lower the confidence threshold from 0.25 to approximately 0.05.



6. Conclusion

GeoVision Pakistan demonstrates that high-quality geospatial deep learning systems for developing-world contexts can be built entirely from freely available satellite data and computing resources. Task A achieved a test mIoU of 0.456 for 7-class LULC segmentation using Swin-UNet, with spatial coherence attributable to the hierarchical self-attention mechanism. Task B achieved near-perfect graveyard detection (mAP50 = 0.995) using YOLOv8s fine-tuned on an OSM-annotated dataset spanning four cities.

Key directions for future work include: (1) applying class-weighted loss and expanding Task A training to the full 11,295 clean tiles to improve Built-up and Wetland IoU; (2) cosine annealing scheduling to approach $\text{mIoU} \geq 0.55$; (3) lowering YOLOv8 confidence threshold to 0.05 for operational use; (4) extending Task B to additional urban facility categories (mosques, schools, hospitals); and (5) reprocessing Task A at 10-metre resolution to eliminate spectral mixing at class boundaries.

References

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Team Member Contributions

Saad Ali — MSDS25066 (Team Leader)

Saad led the overall project architecture and coordinated all phases from inception to submission. His primary contributions were: (1) design and implementation of the complete Google Earth Engine data acquisition pipeline for both tasks; (2) development of the Task A Swin-UNet model architecture, including the 6-to-3 channel projection layer and the custom convolutional decoder with skip connections; (3) implementation of the dataset cleaning pipeline including the NaN audit, orphan removal, dtype casting, and reproducible splits.json generation; (4) conducting all Task A model training experiments across 30 epochs on Google Colab; (5) producing all Task A visualisations (training curves, per-class IoU bar chart, normalised confusion matrix, qualitative prediction grids); and (6) writing the majority of this report, including methodology, results interpretation, and discussion sections.

Numan Hassan — MSDS25067

Numan led the Task B dataset construction pipeline. His contributions were: (1) design and execution of the Overpass API querying workflow to extract graveyard polygons from OpenStreetMap for all four cities; (2) implementation of the GeoTIFF affine transform projection pipeline to convert OSM polygon coordinates to YOLO-format pixel-space bounding boxes; (3) construction of the complete YOLO directory structure (images/train, images/val, images/test, labels/* subdirectories) and the dataset.yaml configuration file; (4) identification and resolution of the stale path bug in dataset.yaml that pointed to the wrong GeoVision exports folder; and (5) validation of label-image correspondence counts across all three splits.

Afzaal Kareem — MSDS25062

Afzal led the Task B model training and evaluation. His contributions were: (1) fine-tuning of the YOLOv8s model on the graveyard detection dataset using the Ultralytics training API; (2) analysis and documentation of the early stopping behaviour (triggered at epoch 38 after best epoch 28); (3) generation and analysis of all Task B diagnostic visualisations including the Precision-Recall curve, F1-Confidence curve, YOLO confusion matrix, and training metric panels; (4) ground truth vs. prediction comparison visualisations for six representative test cities; (5) investigation and documentation of the confidence score calibration issue (Section 5.2.3), identifying it as a known YOLOv8 artefact for single-class high-background-ratio datasets; and (6) compilation of the technical environment appendix.

Usama Umer — MSDS25044

Usama contributed to dataset preparation, quality assurance, and reporting. His contributions were: (1) implementation and execution of the Task A sliding-window tiling pipeline across all four Sentinel-2 GeoTIFF inputs, producing the initial 19,912 tile pool; (2) implementation of the tile validity filters (zero-pixel percentage and mean reflectance threshold checks); (3) execution of the NaN audit sampling and removal pipeline, identifying and removing the 10 affected tiles; (4) Task B city GeoTIFF acquisition and initial tile generation using non-overlapping 256×256 windows with 30% zero-pixel filtering; and (5) proofreading, figure caption writing, and formatting of the final report and README documentation for the GitHub repository.